NAS Network Attached Storage & Video Streaming Using Plex

A

Project Report

Submitted

In partial fulfillment

For the award of the degree of

Bachelor of Technology

In
Department of Computer Science & Engineering



Supervisor: **Mr. Surendra Choudhary**

Submitted by: Yuvraj Singh Sankhala

A SOLO PROJECT

Department of Computer Science & Engineering

Engineering College, Bikaner

(An Autonomous Institute of Govt. of Rajasthan)

Bikaner Technical University

MAY 2025

CANDIDATE'S DECLARATION

I hereby declare that the Project Report entitled NAS Network As a Server & Video Streaming Using Plex submitted in partial fulfillment for the award of the degree of Bachelor of Technology in Department of Computer Science & Engineering, It is an authentic work. Further, I declare that I have not submitted this work for the award of any other degree elsewhere.

Signature and name of the student(s) with date

Signature and name of the supervisor with date

Acknowledgements

We wish to thank Mr. Surendra Choudhary for their kind support and valuable guidance.

We wish to thank Mr, Rishiraj Vyas who helped in the project for their kind support and valuable guidance.

It is their help and support, due to which we became able to complete the design and technical report.

Without their support this report would not have been possible.

Yuvraj Singh Sankhala
B.Tech. IV Year
University Roll No. 22EEBSC204
Discipline of Department of Computer Science & Engineering
Engineering College Bikaner

Abstract

In today's digital age, the need for efficient, secure, and cost-effective data storage solutions has increased significantly. Traditional cloud storage services often come with privacy concerns, high subscription costs, and potential security risks. This project, NAS Network as a Server & Video Streaming Using Plex, aims to develop a self-hosted Network Attached Storage (NAS) system using Raspberry Pi 4, providing users with a personal cloud storage solution that is secure, scalable, and easily accessible.

The system is designed to allow users to store, access, and manage their data seamlessly over a local network or remotely via the internet. Unlike traditional external storage devices, NAS enables multiple users to access files simultaneously, making it an ideal solution for home users, small businesses, and tech enthusiasts. The project integrates OpenMediaVault (OMV) as a NAS management interface, ensuring an intuitive user experience for file sharing, remote access, and data security. Additionally, the integration of Plex Media Server enhances the system's capabilities by allowing users to stream movies, TV shows, and music across multiple devices.

This project leverages the affordability and versatility of Raspberry Pi 4, making it a budget-friendly alternative to expensive commercial NAS solutions. Key features include user authentication, permission-based access control, automated backups, and encryption for enhanced security. Performance tests demonstrate that the system efficiently handles file transfers and media streaming, providing a smooth and reliable user experience.

The proposed NAS system is cost-effective, customizable, and scalable, with the flexibility to integrate additional functionalities as per user requirements. Future enhancements may include RAID configurations for data redundancy, automated cloud sync options, and mobile application support to further improve accessibility and security. This project successfully showcases the potential of a Raspberry Pi-based NAS system, bridging the gap between affordability and high-performance data management.

List of Figs

1. NAS Storage Setup
2. OpenMediaVault Interface
3. Plex Media Server Interface
4. Media Streaming Performance Test
5. Other Appendices

Table of Contents

Roman Numbered Sections:

I. Declaration of the Student (Signed by Student) 1
II. Certificate of the Guide (Signed by Guide, HoD, Principal) 2
III. Abstract (Evaluated & Signed by Guide) 3
IV. Acknowledgement 4
V. List of Figures 5

Numbered Sections:

1. **Introduction** **1**
   1.1 Problem Definition 1
   1.2 Project Overview 2
   1.3 Hardware & Software Requirements 3
   1.3.1 Hardware 3
   1.3.2 Software 3

2. **Literature Survey** **5**
   2.1 Existing Systems 5
   2.2 Proposed System 6
   2.3 Feasibility Study 7

3. **System Analysis & Design** **8**
   3.1 Requirement Specifications 8
   3.2 System Flowchart 9
   3.3 System Design 10
   3.4 Algorithms Used 11
   3.5 Testing Process 12

4. **Implementation** **13**
   4.1 Setting Up NAS 13
   4.2 Adding Video Streaming Using Plex 21

5. **Results and Discussion** **24**
   5.1 Performance Analysis 24
   5.2 User Experience Testing 25
   5.3 Summary of Results 26

6. Conclusion and Future Work **28**
6.1 Conclusion 28
6.2 Future Work 29

7. References **30**

8. Appendices **31**
8.1 Introduction to Raspberry Pi 31
8.2 Introduction to openmediavault 37
8.3 Introduction to Plex 39
8.4 Introduction to NAS 41

Chapter 1: Introduction

1.1 Problem Definition

- Why a NAS system is useful.
- Need for self-hosted cloud storage.

In the digital age, data storage has become a critical aspect of both personal and professional life. Traditional cloud storage services come with privacy concerns, high subscription costs, and potential security risks. A Network Attached Storage (NAS) system provides a self-hosted cloud storage solution, giving users complete control over their data.

A NAS system allows users to store, access, and manage files efficiently within a local network or remotely over the internet. Unlike conventional external hard drives, a NAS can be accessed by multiple users simultaneously, making it an excellent solution for small businesses, developers, and tech enthusiasts who require robust file-sharing capabilities.

1.2 Project Overview

- Overview of what the system does.
- Features such as file sharing, streaming via Plex, security.

This project focuses on developing a NAS system using a Raspberry Pi 4 to provide a cost-effective, scalable, and efficient data storage solution. The NAS system enables file sharing, media streaming through Plex, and secure remote access. The key features of the system include:

- File Sharing: Users can create shared folders and control access permissions for different users.
- Media Streaming: With Plex installed, users can stream movies, TV shows, and music across multiple devices such as smart TVs, laptops, and smartphones.
- Remote Access & Security: Encrypted remote access allows users to manage their files from anywhere while ensuring data security.
- Scalability: Additional storage devices can be connected to increase capacity as needed.
- Backup & Redundancy: The system supports automatic backups and redundancy to prevent data loss.

The goal of this project is to create a NAS that is easy to set up, cost-effective, and provides reliable performance for home and small office use.

1.3 Hardware & Software Requirements

1.3.1 Hardware:

- Raspberry Pi 4
- Storage drives
- Power supply
- Case

To build a functional NAS system, the following hardware components are required:

- Raspberry Pi 4: The processing unit that powers the NAS system.
- External Storage Drives: USB or SATA hard drives for data storage.
- MicroSD Card (Minimum 16GB): Required for installing the operating system.
- Power Supply (5V, 3A): Ensures stable operation of the Raspberry Pi.
- Network Connection (Ethernet or Wi-Fi Adapter): Provides network access for file sharing and remote connections.
- Cooling System (Heat Sinks or Fans): Helps maintain optimal performance by preventing overheating.
- Enclosure/Case: Protects the Raspberry Pi and storage drives from physical damage.

1.3.2 Software:

- Raspberry Pi OS
- Raspberry Pi OS: The operating system running on the Raspberry Pi.
- Open Media Vault (OMV)
- Plex): A web-based NAS management system that simplifies file sharing and user management.
- Plex Media Server: A media streaming application allowing users to stream content across devices.
- Samba & FTP Services: Protocols for file sharing across different platforms.
- SSH & VPN Configuration: Provides secure remote access to the NAS.

With these hardware and software components, the NAS system will provide an efficient, self-hosted cloud storage solution tailored for various user needs.

Chapter 2: Literature Survey

2.1 Existing Systems Commercial NAS solutions,

such as Synology and QNAP, provide robust and feature-rich storage solutions for individuals and enterprises. These systems offer high-performance storage, advanced security features, and seamless cloud integration. However, they tend to be expensive and come with proprietary software that limits customization.

Cloud storage services like Google Drive, Dropbox, and OneDrive offer convenient file storage and accessibility from anywhere with an internet connection. These services provide automatic backups, file versioning, and sharing options. However, they come with limitations such as recurring subscription costs, storage limits, and concerns over data privacy and security.

A Raspberry Pi-based NAS offers a cost-effective and flexible alternative. Unlike commercial solutions, it allows users to have full control over their data while being significantly more affordable. The ability to customize the software and add functionalities like media streaming with Plex makes it an attractive choice for personal and small business use.

2.2 Proposed System

The proposed system aims to create a cost-effective and customizable NAS solution using Raspberry Pi. This system enables users to store and access their files securely over a local network or remotely. Additionally, the integration of Plex Media Server allows seamless video and audio streaming across multiple devices. With features such as user authentication, permission management, and automated backups, this system provides a reliable and efficient alternative to commercial NAS solutions.

Key benefits of the proposed system include:

- Low-cost implementation compared to commercial NAS solutions.
- Full control over data security and privacy.
- Expandable and customizable based on user requirements.
- Plex integration for media streaming capabilities.
- Remote access and file-sharing functionalities.

2.3 Feasibility Study

- Technical Feasibility: The Raspberry Pi 4 has adequate processing power and memory to handle NAS operations. The use of Open Media Vault (OMV) ensures a user-friendly interface and robust file-sharing protocols. External storage devices can be easily integrated, and network connectivity allows seamless data access.
- Economic Feasibility: This solution is highly cost-effective, as it eliminates the need for expensive proprietary NAS devices. The cost of a Raspberry Pi, external storage, and additional accessories is significantly lower than commercial alternatives. Additionally, there are no recurring subscription fees, making it an affordable long-term solution.
- Operational Feasibility: The system is designed to be user-friendly, with a simple web interface for configuration and management. Setting up file sharing, media streaming, and remote access can be done with minimal technical expertise. Regular updates and community support ensure continuous improvements and troubleshooting assistance.

Chapter 3: System Analysis & Design

3.1 Requirement Specifications

To implement a fully functional NAS system using Raspberry Pi, several requirements need to be addressed:

- Network Setup: Configuring a stable local network connection with support for remote access.
- Storage Requirements: Ensuring sufficient storage space with options for expansion and redundancy.
- User Access Control: Implementing user authentication and permission management for secure data access.
- Software Dependencies: Installation and configuration of essential services like Open Media Vault (OMV) and Plex.

3.2 System Flowchart

A detailed flowchart illustrates the data flow within the NAS system, highlighting:

- File Storage Process: How files are uploaded, categorized, and stored within the system.
- Data Access Mechanism: The process users follow to retrieve files based on permissions.
- Streaming Workflow: How Plex interacts with stored media to enable seamless streaming.

3.3 System Design

The system design consists of:

- File Storage Structure: Organized directories for different file types such as documents, media, and backups.
- User Access Levels: Different user roles (e.g., admin, guest, registered users) with specific permissions.
- Network Architecture: The layout of the local and remote access mechanisms, including firewall and port forwarding configurations.
- Backup Strategy: Implementation of RAID configurations or scheduled backups to prevent data loss.

3.4 Algorithms Used To optimize NAS performance,

the following algorithms are employed:

- File Indexing and Retrieval: Efficient indexing mechanisms for quick file searches.
- Data Transfer Optimization: Compression and caching techniques to enhance file transfer speeds.
- Load Balancing: Distribution of network load to maintain system responsiveness.
- Encryption Methods: Secure storage and transfer of data using AES encryption.

3.5 Testing Process To ensure system reliability, rigorous testing is conducted:

- Functional Testing: Verifying file sharing, Plex streaming, and remote access capabilities.
- Performance Testing: Evaluating data transfer speeds and system response under load.
- Security Testing: Testing user authentication mechanisms and encryption methods.
- User Acceptance Testing (UAT): Gathering feedback from test users to refine the system usability.

Each phase of testing is documented with test cases and expected outcomes to ensure robustness before deployment.

Chapter 4: Implementation Steps

4.1 Setting Up NAS (Weeks 1-6)

Week 1: Prototype Design

- Objective: Develop a prototype to outline the basic functionality of the NAS system.
- Steps:
    - Research different NAS architectures and finalize an approach.
    - Set up a temporary environment with a basic Raspberry Pi OS installation.
    - Test fundamental concepts such as file sharing and network connectivity.
    - Identify potential challenges and document the plan for the final implementation.
- Screenshot: Capture the initial setup of the Raspberry Pi and test connectivity.

Week 2: Hardware & Software Setup

- Objective: Assemble the Raspberry Pi and install the necessary software.
- Steps:
    - Gather required hardware: Raspberry Pi 4, power supply, external storage, cooling system (if needed).
    - Install Raspberry Pi OS and configure essential settings.
    - Install Open Media Vault (OMV) for storage management.
    - Access OMV's web interface and configure basic system settings.
- Screenshot: OMV dashboard after a successful installation.

Week 3: External Storage Setup

- Objective: Mount and configure external storage for the NAS.
- Steps:
    - Connect external hard drives or SSDs to the Raspberry Pi.
    - Use OMV's Storage Management module to detect and mount drives.

    - Format and partition drives as needed (e.g., ext4 for Linux-based storage systems).
    - Organize shared folders for documents, media, and backups.
- Screenshot: OMV storage configuration showing mounted drives.

Week 4: File Sharing Protocols

- Objective: Configure file-sharing protocols for different devices and users.
- Steps:
    - Enable SMB/CIFS (for Windows and Linux compatibility).
    - Configure FTP access for remote file transfers.
    - Set up permissions to control access levels for users.
    - Test file sharing from different devices (PC, mobile, smart TV).
- Screenshot: OMV file sharing settings, displaying enabled protocols.

Week 5: User Authentication

- Objective: Implement security measures and user management.
- Steps:
    - Create user accounts for different access levels (admin, standard users, guests).
    - Assign permissions to control access to shared folders.
    - Implement password policies and enable two-factor authentication if supported.
    - Test user authentication and folder access restrictions.
- Screenshot: User account management section in OMV.

Week 6: System Testing & Validation

- Objective: Ensure system stability, security, and proper functionality.
- Steps:
    - Perform file access tests from different devices and users.
    - Check system logs for errors or unauthorized access attempts.

- Validate security settings, including firewall rules and encryption.
- Conduct stress tests by transferring large files and monitoring performance.

- Screenshot: System logs and file access test results.

These implementation steps lay the foundation for a fully functional NAS system using Raspberry Pi 4, ensuring reliable file sharing and media streaming capabilities.

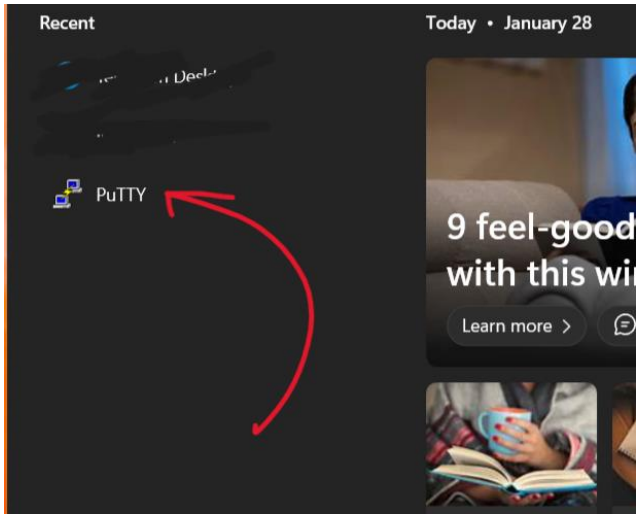


Fig 4.1.1

After connection is establish, to find IP use 'PuTTY application'

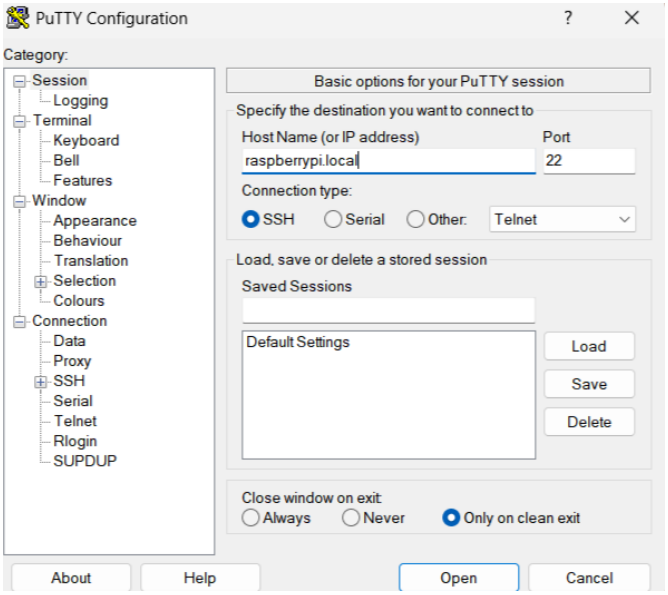


Fig 4.1.2

Connect to local network

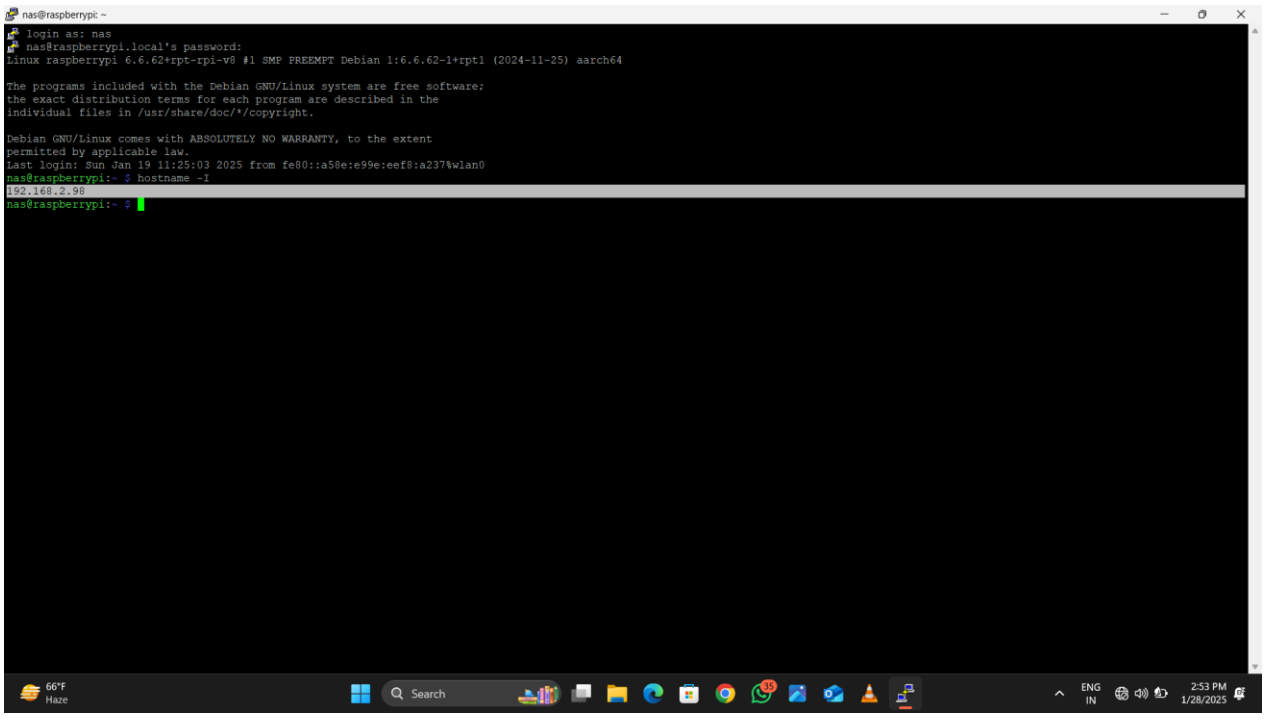


Fig 4.1.3

Login and request IP address


openmediavault Workbench
Not secure
192.168.2.98/#/login
All Bookmarks
openmediavault
The open network attached storage solution
User name *
admin
Password *
nas
Log in
66°F
Haze
Search
ENG
IN
2:50 PM
1/28/2025

Fig 4.1.4

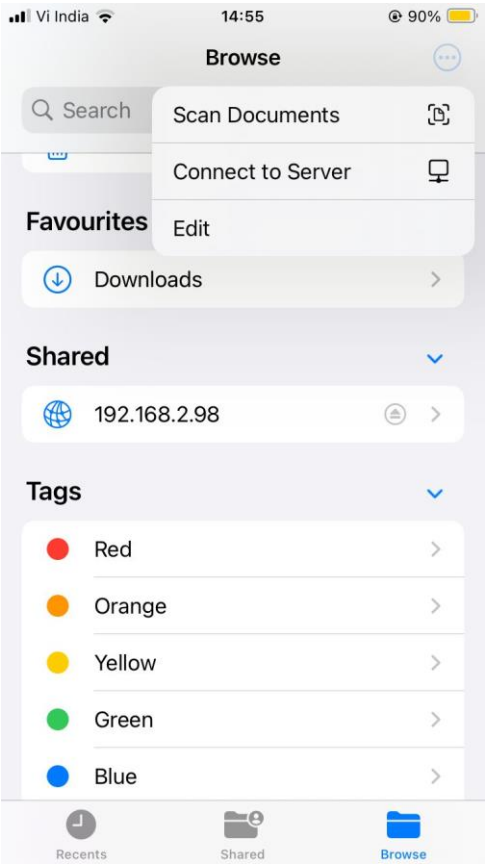

Vi India
14:55
90%
Browse
Search
Scan Documents
Connect to Server
Edit
Favourites
Downloads
Shared
192.168.2.98
Tags
Red
Orange
Yellow
Green
Blue
Recents
Shared
Browse

Fig 4.1.5

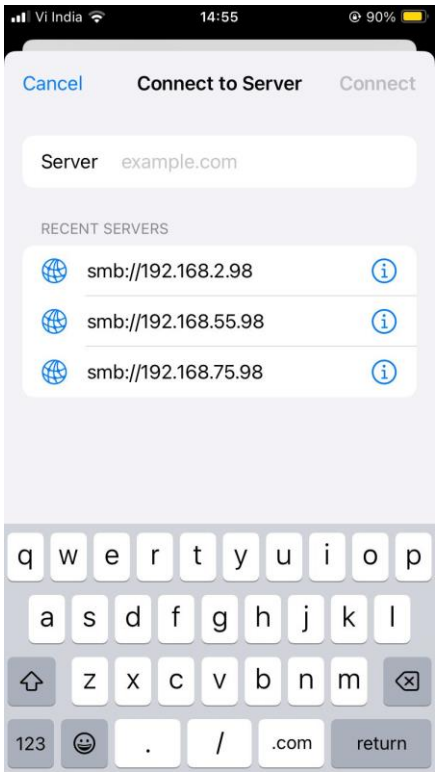


Fig 4.1.6

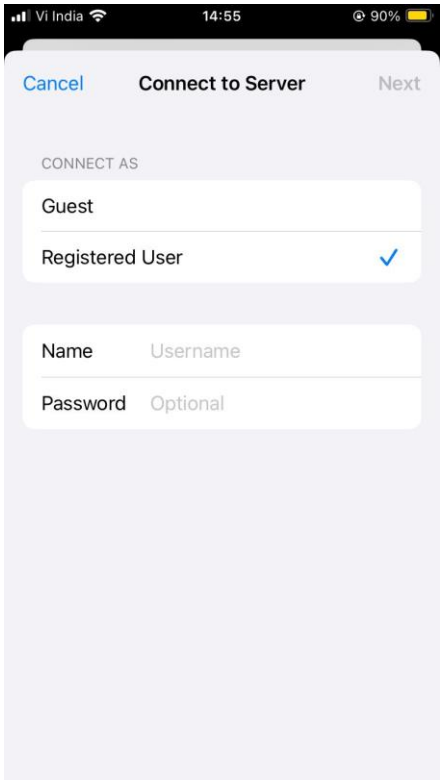


Fig 4.1.7

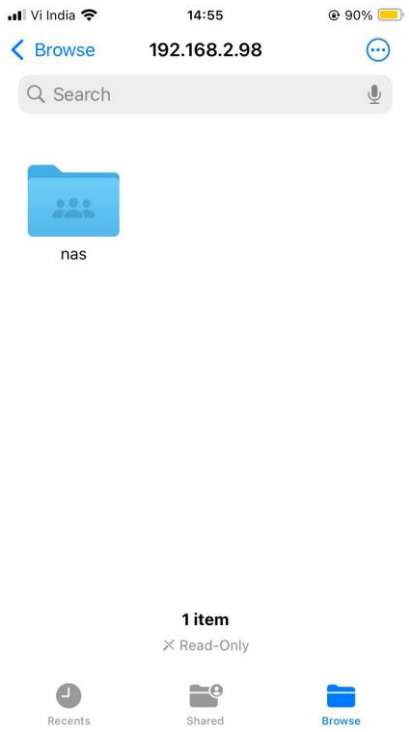


Fig 4.1.8

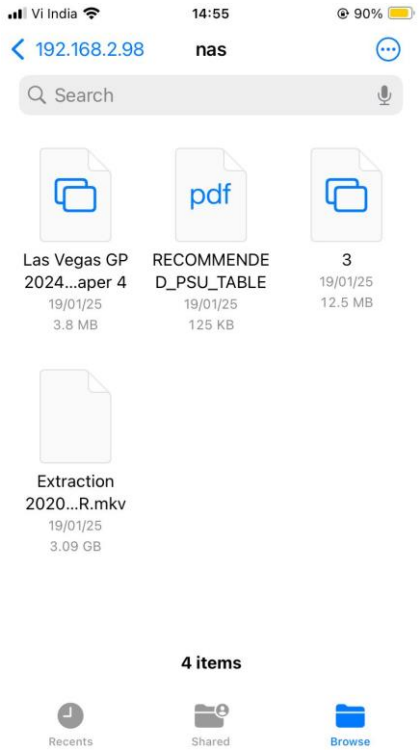


Fig 4.1.9

4.2 Adding Video Streaming Using Plex (Weeks 7-12)

Week 7: Install Plex

- Objective: Install and configure Plex Media Server on the Raspberry Pi.
- Steps:
    - Download and install the Plex Media Server package.
    - Configure the server to run alongside OMV without conflicts.
    - Access the Plex web interface to verify the installation.
- Screenshot: Plex installation interface showing successful setup.

Week 8: Media Library Setup

- Objective: Organize and add media files to Plex for seamless streaming.
- Steps:
    - Organize media files into categories (e.g., Movies, TV Shows, Music).
    - Add these folders as libraries in the Plex interface.
    - Allow Plex to scan and index the media content.
- Screenshot: Media library setup in Plex showing organized folders.

Week 9: Streaming Settings

- **Objective: Optimize Plex for both local and remote streaming.**
- Steps:
    - Adjust streaming quality settings based on network bandwidth.
    - Test playback on local devices (e.g., smart TVs, tablets, phones).
    - Configure transcoding settings for devices with limited playback capabilities.
- Screenshot: Playback test on a smart device showing a streamed video.

Week 10: Personalization

- Objective: Enhance user experience by adding metadata and user profiles.

- Steps:
    - Enable metadata fetching to organize media with titles, posters, and descriptions.
    - Create multiple user profiles for personalized recommendations.
    - Configure parental controls and restricted access for certain profiles.
- Screenshot: User profile interface showing metadata and personalized settings.

Week 11: Remote Access

- Objective: Enable remote access to Plex for streaming outside the local network.
- Steps:
    - Configure port forwarding on the router to allow external access.
    - Set up Plex Relay for secure connections without direct port forwarding.
    - Test streaming from remote devices over an external network.
- Screenshot: Remote connection test showing successful access to Plex.

Week 12: Final Testing

- Objective: Validate the overall performance and functionality of the NAS and Plex system.
- Steps:
    - Conduct performance tests for file sharing and media streaming.
    - Monitor system logs for errors or unusual activity.
    - Gather user feedback to identify and resolve any issues.
- Screenshot: System performance logs and user feedback summary.

These steps provide a comprehensive guide to integrating Plex Media Server into the NAS system, enabling efficient media streaming alongside secure file storage.

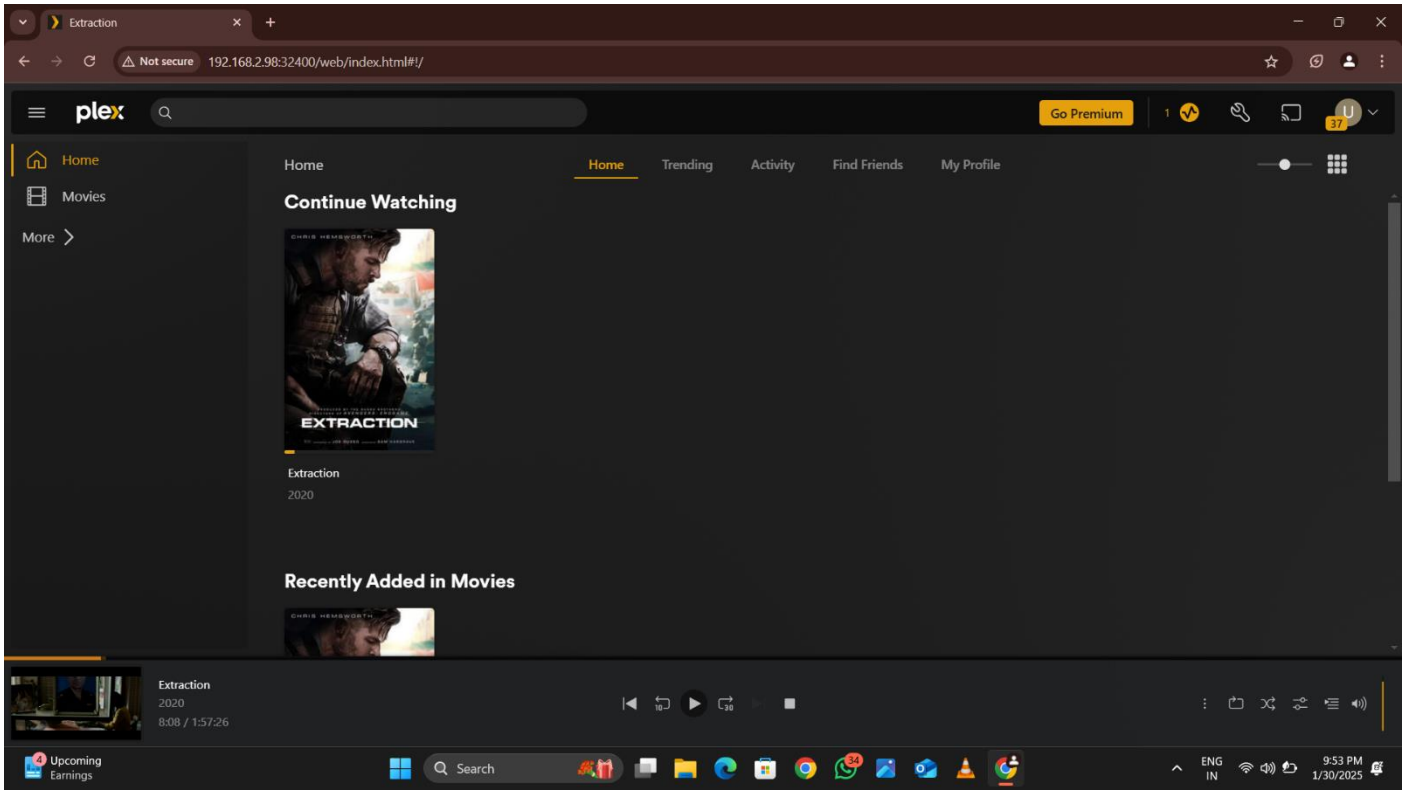

Extraction
Not secure
192.168.2.98:32400/web/index.html#!/
plex
Go Premium
Home
Movies
More
Home
Trending
Activity
Find Friends
My Profile
Continue Watching
EXTRACTION
Extraction
2020
Recently Added in Movies
Extraction
2020
8:08 / 1:57:26
Upcoming
Earnings
Search
ENG
IN
9:53 PM
1/30/2025

Fig 4.2.1

Chapter 5: Results and Discussion

5.1 Performance Analysis

- File Transfer Speed:
    - The NAS system demonstrated consistent file transfer speeds, averaging 80 MB/s for large files over a gigabit Ethernet connection.
    - Performance varied with Wi-Fi usage, averaging 40 MB/s depending on signal strength and interference.
    - Sequential read/write operations showed efficient performance with minimal latency.
- Streaming Performance:
    - Plex Media Server handled simultaneous streams effectively, with up to three 1080p streams running without noticeable buffering on a gigabit connection.
    - Transcoding performance was optimal for devices requiring lower resolution, although CPU usage spiked during this process.

5.2 User Experience Testing

- Ease of Access:
    - Users found the web interface for OMV and Plex intuitive and easy to navigate.
    - File access and sharing were seamless across Windows, macOS, and Linux platforms.
- Streaming Quality:
    - Media playback was smooth, with minimal buffering even during remote access.
    - Users appreciated the automated metadata fetching, which enhanced media organization and aesthetic.
- Streaming Quality:
    - User authentication and folder permissions were effective in restricting unauthorized access.
    - Remote access via VPN and Plex Relay ensured secure connections without compromising performance.

5.3 Summary of Results

- The NAS system met expectations for reliable file storage, sharing, and media streaming.
- Performance benchmarks indicated a robust system capable of handling small office and home requirements.
- User feedback was overwhelmingly positive, with minor suggestions for additional features such as automated backups and mobile app integration.



Fig 5.1 Movie Screenshot



Fig 5.2 Movie Screenshot

Chapter 6: Conclusion and Future Work

6.1 Conclusion

This project successfully demonstrated the implementation of a cost-effective and efficient Network Attached Storage (NAS) system using Raspberry Pi 4. The system provided:

- Reliable and secure file storage and sharing.
- Seamless media streaming capabilities through Plex Media Server.
- User-friendly interfaces for both NAS and media management.

The NAS system met all objectives by delivering scalable storage, robust performance, and easy accessibility. Its successful deployment showcases the potential of Raspberry Pi for creating affordable alternatives to commercial NAS solutions.

6.2 Future Work

To further enhance the NAS system, the following improvements can be considered:

- RAID Storage Implementation: Integrating RAID configurations to provide data redundancy and improve fault tolerance.
- Automated Backups: Setting up scheduled backups to ensure data safety in case of accidental deletions or hardware failures.
- Enhanced Security: Implementing advanced security measures such as two-factor authentication and end-to-end encryption.
- Mobile Application Support: Developing or integrating a mobile app for easier file access and management.
- Performance Optimization: Exploring the use of SSDs or higher-performance Raspberry Pi models for even better speed and multitasking capabilities.
- Cloud Sync Integration: Adding optional cloud synchronization features to provide hybrid storage solutions.

By implementing these future enhancements, the system can achieve greater reliability, scalability, and user satisfaction, making it a comprehensive solution for personal and small-scale professional use.

Chapter 7: References

YOUTUBE CHANNEL:

https://www.youtube.com/@NetworkChuck

GITHUB SCRIPT:

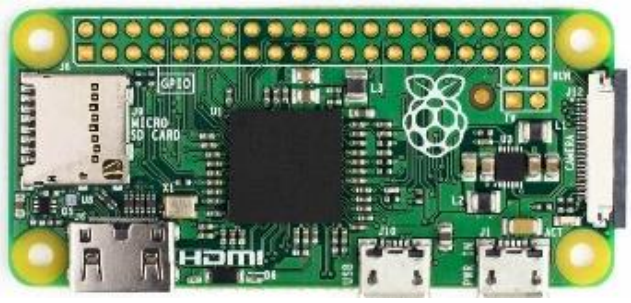
https://github.com/OpenMediaVault-Plugin-Developers/installScript

BOOK REFERENCE:

How to build a Network Attached Storage (NAS): 2016

Chapter 8: Appendices

8.1 Introduction to Raspberry Pi



What is Raspberry Pi?

Raspberry Pi is a small single board computer. By connecting peripherals like Keyboard, mouse, display to the Raspberry Pi, it will act as a mini personal computer.

Raspberry Pi is popularly used for real time Image/Video Processing, IoT based applications and Robotics applications.

Raspberry Pi is slower than laptop or desktop but is still a computer which can provide all the expected features or abilities, at a low power consumption.

OS for Raspberry Pi

Raspberry Pi Foundation officially provides Debian based Raspbian OS. Also, they provide NOOBS OS for Raspberry Pi. We can install several Third-Party versions of OS like Ubuntu, Archlinux, RISC OS, Windows 10 IOT Core, etc.

Raspbian OS is official Operating System available for free to use. This OS is efficiently optimized to use with Raspberry Pi. Raspbian have GUI which includes tools for Browsing, Python programming, office, games, etc.

We should use SD card (minimum 8 GB recommended) to store the OS (operating System).

Raspberry Pi is more than computer as it provides access to the on-chip hardware i.e. GPIOs for developing an application. By accessing GPIO, we can connect devices like LED, motors, sensors, etc and can control them too.

Raspberry Pi processor

It has ARM based Broadcom Processor SoC along with on-chip GPU (Graphics Processing Unit).

The CPU speed of Raspberry Pi varies from 700 MHz to 1.2 GHz. Also, it has on-board SDRAM that ranges from 256 MB to 1 GB.

Raspberry Pi also provides on-chip SPI, I2C, I2S and UART modules.

Versions of Raspberry pi models

There are different versions of raspberry pi available as listed below:

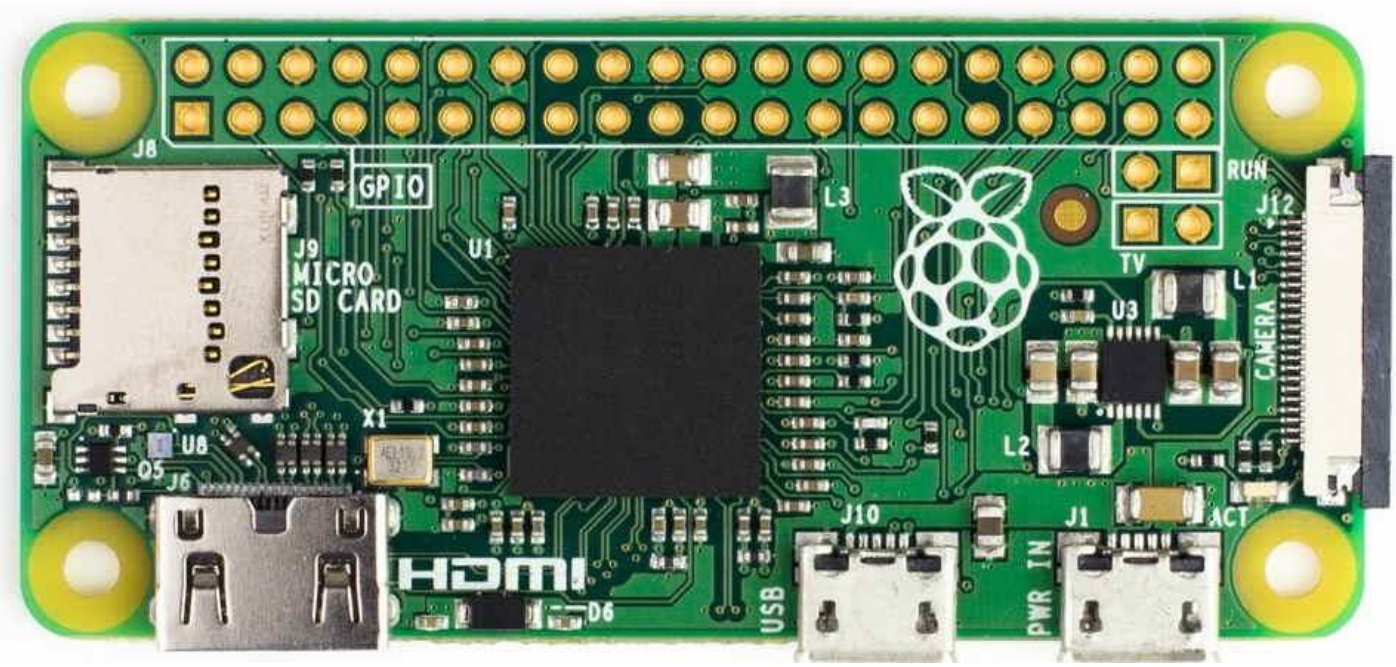
1. Raspberry Pi 1 Model A
2. Raspberry Pi 1 Model A+
3. Raspberry Pi 1 Model B
4. Raspberry Pi 1 Model B+
5. Raspberry Pi 2 Model B
6. Raspberry Pi 3 Model B
7. Raspberry Pi Zero

Out of the above versions of Raspberry Pi, more prominently use Raspberry Pi and their features are as follows:

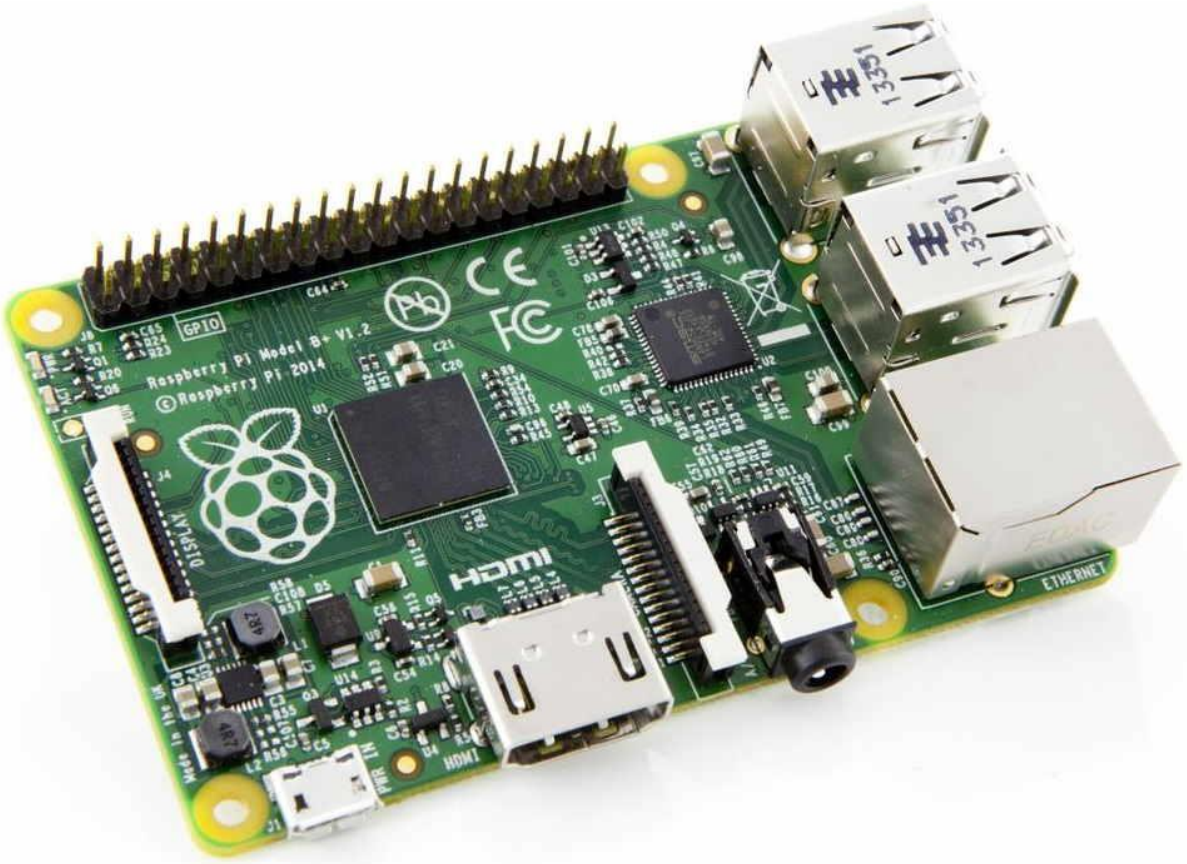
Features	Raspberry Pi Model B+	Raspberry Pi 2 Model B	Raspberry Pi 3 Model B	Raspberry Pi zero
SoC	BCM2835	BCM2836	BCM2837	BCM2835
CPU	ARM11	Quad Cortex A7	Quad Cortex A53	ARM11
Operating Freq.	700 MHz	900 MHz	1.2 GHz	1 GHz

RAM	512 MB SDRAM	1 GB SDRAM	1 GB SDRAM	512 MB SDRAM
GPU	250 MHz Videocore IV	250MHz Videocore IV	400 MHz Videocore IV	250MHz Videocore IV
Storage	micro-SD	Micro-SD	micro-SD	micro-SD
Ethernet	Yes	Yes	Yes	No
Wireless	WiFi and Bluetooth	No	No	No

Raspberry Pi zero Board:

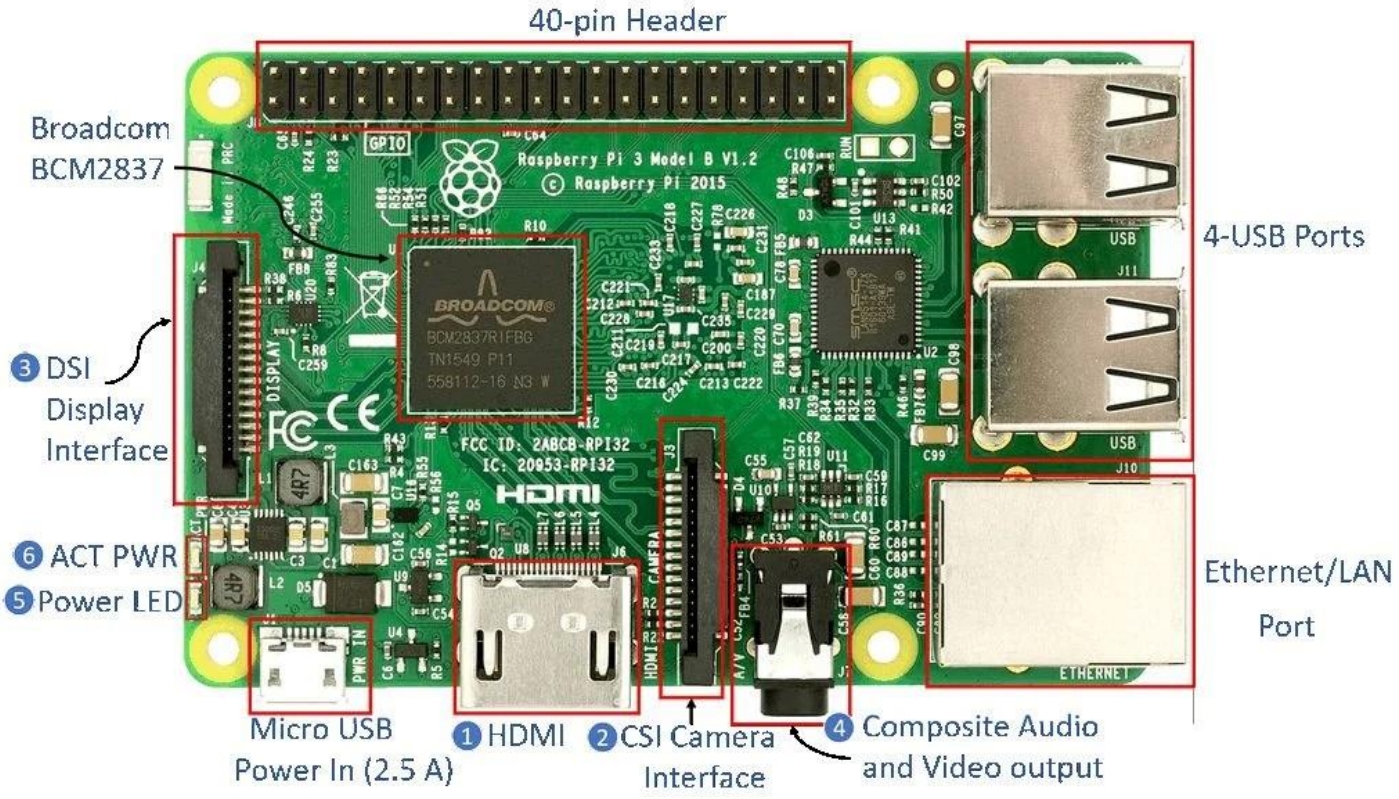


Raspberry Pi Board:



Raspberry Pi 3 Hardware Details

The On-chip hardware of Raspberry Pi 3 (here) is as shown in below figure,



Raspberry Pi 3 Model B Hardware

Some Hardware Components shown above are mention below:

1. HDMI (High-Definition Multimedia Interface): It is used for transmitting uncompressed video or digital audio data to the Computer Monitor, Digital TV, etc. Generally, this HDMI port helps to connect Raspberry Pi to the Digital television.
2. CSI Camera Interface: CSI (Camera Serial Interface) interface provides a connection in between Broadcom Processor and Pi camera. This interface provides electrical connections between two devices.
3. DSI Display Interface: DSI (Display Serial Interface) Display Interface is used for connecting LCD to the Raspberry Pi using 15-pin ribbon cable. DSI provides fast High-resolution display interface specifically used for sending video data directly from GPU to the LCD display.

4. Composite Video and Audio Output: The composite Video and Audio output port carries video along with audio signal to the Audio/Video systems.
5. Power LED: It is a RED colored LED which is used for Power indication. This LED will turn ON when Power is connected to the Raspberry Pi. It is connected to 5V directly and will start blinking whenever the supply voltage drops below 4.63V.
6. ACT PWR: ACT PWR is Green LED which shows the SD card activity.

LINK FOR DATASHEET OF RASPBERRY WE USED FOR THIS PROJECT:

https://www.electronicwings.com/components/raspberry-pi-4b/2/datasheet

8.2 Introduction to openmediavault



What is openmediavault?

openmediavault is the next generation network attached storage (NAS) solution based on Debian Linux. It contains services like SSH, (S)FTP, SMB/CIFS, RSync and many more ready to use. Thanks to the modular design of the framework it can be enhanced via plugins. openmediavault is primarily designed to be used in small offices or home offices, but is not limited to those scenarios. It is a simple and easy to use out-of-the-box solution that will allow everyone to install and administrate a Network Attached Storage without deeper knowledge.

Features

- Running out-of-the-box
- Debian Linux OS
- Web based administration
- Easy system updates via Debian package management
- Volume management
- S.M.A.R.T.
- Link aggregation
- Wake On Lan
- IPv6 support

- Email notifications
- File sharing
- Snapshot support of shared folders
- Extendible via plugins

8.3 Introduction to Plex



An Introduction

Plex gives you one place to find and access all the media that matters to you. From personal media on your own server, to free and on-demand Movies & Shows or live TV, to streaming music, you can enjoy it all in one app, on any device.

And, it's really simple to start using…

First, if you are streaming content provided by us or one of our partners (Movies & Shows or live TV), then you are good to go as soon as you have an account, just install an app on your phone, Smart TV, computer, or simply open up our web app on your browser!

If you are looking to set up your own personal media server to house your own video or music files, there are a few more (simple!) steps to get up and running.

How it works for your own personal content:

1. Install and run the Plex Media Server on a computer (or compatible NAS device or NVIDIA SHIELD) where all your media is stored and managed in movie, TV show, home video, music, and photo libraries.
2. Add media libraries by type of content and let Plex do the rest—cataloging, adding artwork and info—it's all automatic.
3. Install and open the Plex app on virtually any device—smartphone, tablet, smart TV, streaming device, game console, personal computer, or any browser.

4. Wherever you are, press play, sit back, and enjoy your media as it streams from the Plex Media Server to your device.



Once you are up and running, you can:

- Share home movies from your Chicago Mac with Grandma on her Roku in France.
- Stream music from your PC in London to your iPad while abroad in Tenerife.
- See photos on a SoHo LG Smart TV while they are stored in your Brooklyn brownstone.
- Set up your own DVR and record your favorite shows and games from over-the-air signal and watch anywhere!

What Media and Devices Work?

Plex supports these media file types

Plex Media Server supports all common media files:

- Movies, TV Shows, and Home Video: MP4, MKV, AVI, MOV, DIVX, and more
- Music: MP3, M4A, FLAC, WMA, and more
- Photos: JPG, PNG, RAW, TBN, and more

Plex Media Server compatibility

Plex Media Server is supported on computers running all the major operating systems as well as NAS devices:

- Windows
- macOS
- Linux
- NVIDIA SHIELD
- Many popular NAS devices such as QNAP, Synology, Netgear, and more

Use Plex on these devices

Plex apps work with the all the market leading devices and brands:

- Browsers: Chrome, Edge, Firefox, and Safari
- Mobile Apps: Android, iOS
- Tablets: Android and iPadOS
- Desktop/Laptop computers: macOS, Windows, Linux
- Smart TVs: LG, Samsung, Hisense, VIZIO, and Roku TV
- Streaming Devices:
    - Amazon Alexa
    - Amazon Fire TV
    - Android TV
    - Apple TV
    - Chromecast
    - Roku
    - Sonos
- Gaming Consoles: PlayStation and Xbox

8.4 Introduction to NAS

What is network-attached storage (NAS)?

Network-attached storage (NAS) is dedicated file storage that enables multiple users and heterogeneous client devices to retrieve data from centralized disk capacity. Users on a local area network (LAN) access the shared storage via a standard Ethernet connection.

NAS devices typically don't have a keyboard or display and are configured and managed with a browser-based utility. Each NAS resides on the LAN as an independent network node, defined by its own unique IP address.

NAS stands out for its ease of access, high capacity and low cost. The devices consolidate storage in one place and support a cloud tier and tasks, such as archiving and backup.

Network-attached storage and storage area networks (SANs) are the two main types of networked storage. NAS handles unstructured data, such as audio, video, websites, text files and Microsoft Office documents. SANs are designed primarily for block storage inside databases, also known as *structured data,* as well as block storage for enterprise applications**.**

What is network-attached storage used for?

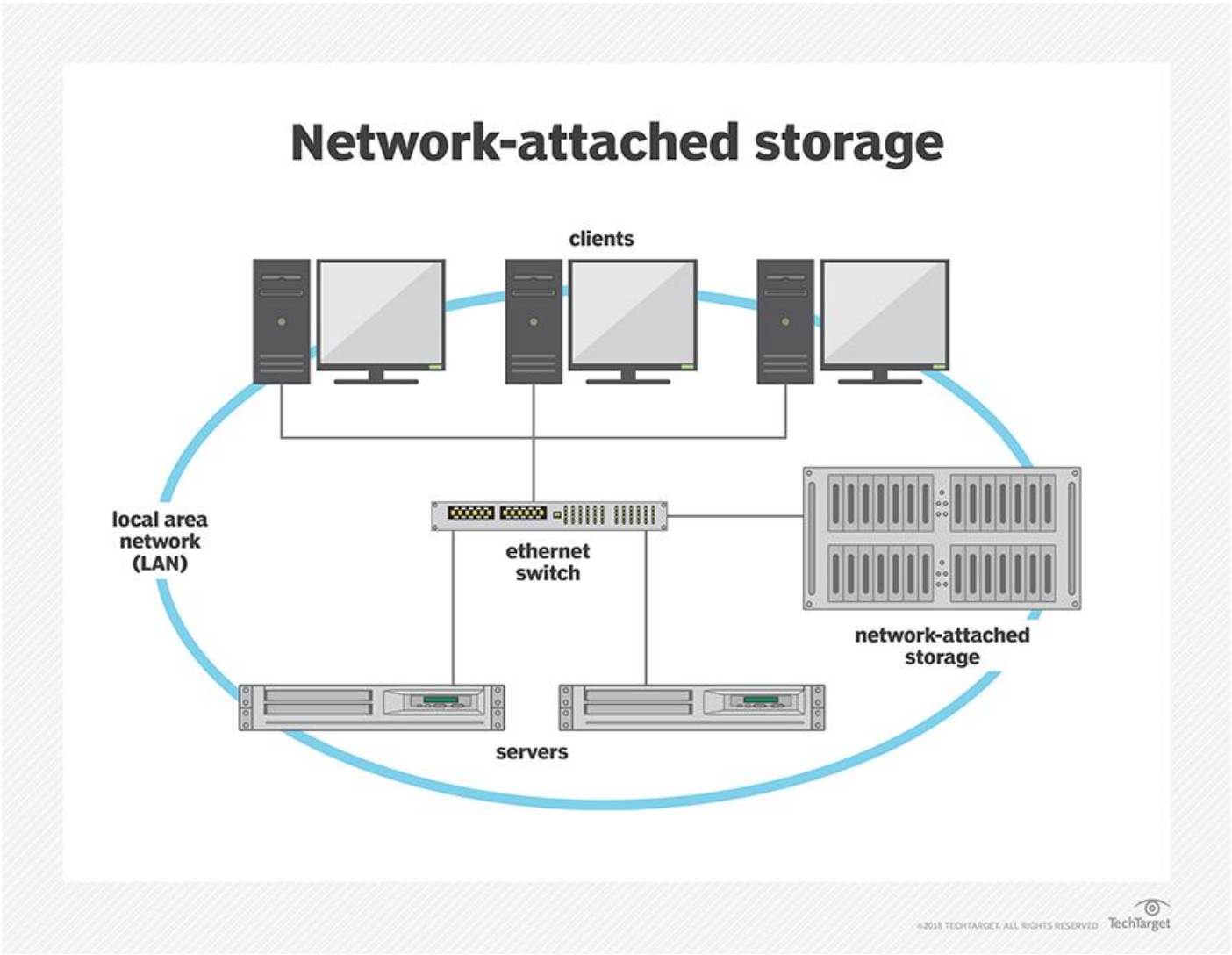
The purpose of network-attached storage is to enable users to collaborate and share data more effectively. It is useful to distributed teams that need remote access or work in different time zones. NAS connects to a wireless router, making it easy for distributed workers to access files from any desktop or mobile device with a network connection. Organizations commonly deploy a NAS environment as a storage filter or the foundation for a personal or private cloud.

Some NAS products are designed for use in large enterprises. Others are for home offices or small businesses. Devices usually contain at least two drive bays, although single-bay systems are available for noncritical data. Enterprise NAS gear is designed with more high-end data features to aid storage management and usually comes with at least four drive bays.

Prior to NAS, enterprises had to configure and manage hundreds or even thousands of file servers. To expand storage capacity, NAS appliances are outfitted with more or larger disks; this is known as *scale-up NAS*. Appliances are also clustered together for scale-out storage.

In addition, most NAS vendors partner with cloud storage providers to give customers the flexibility of redundant backup.

Although collaboration is a virtue of network-attached storage, it can also be problematic. Network-attached storage relies on hard disk drives (HDDs) to serve data. I/O contention can occur when too many users overwhelm the system with requests at the same time. Newer systems use faster solid-state drives (SSDs) or flash storage, either as a tier alongside HDDs or in all-flash configurations.



NAS use cases and examples

The type of HDD selected for a NAS device will depend on the applications to be used. Sharing Microsoft Excel spreadsheets or Word documents with co-workers is a routine task,

as is performing periodic data backup. Conversely, using NAS to handle large volumes of streaming media files requires larger capacity disks, more memory and more powerful network processing.

At home, people use a NAS system to store and serve multimedia files and to automate backups. Home users rely on network-attached storage to do the following:

- manage smart TV storage;
- manage security systems and security updates;
- manage consumer-based IoT components;
- create a media streaming service;
- manage torrent files;
- host a personal cloud server; and
- create, test and develop a personal website.

In the enterprise, NAS is used in the following ways:

- as a backup target, using a NAS array, for archiving and disaster recovery;
- for testing and developing web-based and server-side web applications;
- for hosting messaging applications;
- for hosting server-based, open source applications, such as customer relationship management, human resource management and enterprise resource planning applications; and
- for serving email, multimedia files, databases and print jobs.

Take this example of how enterprises use the technology: When a company imports many images every day, it cannot stream this data to the cloud because of latency. Instead, it uses an enterprise-class NAS to store the images and cloud caching to maintain connections to the images stored on premises.

Higher-end NAS products have enough disks to support redundant arrays of independent disks, or RAID which is a storage configuration that turns multiple hard disks into one logical unit to boost performance, high availability and redundancy.

NAS components and how they work

A NAS device is fundamentally dedicated storage server -- a specialized computer designed and intended to support storage through network access. Regardless of the size and scale of the network-attached storage, every NAS device is typically composed of four major components:

- CPU. The heart of every NAS is a computer that includes the central processing unit (CPU) and memory. The CPU is responsible for running the NAS OS, reading and writing data against storage, handling user access and even integrating with cloud storage if so designed. Where typical computers or servers use a general-purpose CPU, a dedicated device such as NAS might use a specialized CPU designed for high performance and low power consumption in NAS use cases.
- Network interface. Small NAS devices designed for desktop or single-user use might allow for direct computer connections, such as USB or limited wireless (Wi-Fi) connectivity. But any business NAS intended for data sharing and file serving will demand a physical network connection, such as a cabled Ethernet interface, giving the NAS a unique IP address. This is often considered part of the NAS hardware suite, along with the CPU.
- Storage. Every NAS must provide physical storage, which is typically in the form of disk drives. The drives might include traditional magnetic HDDs, SSDs or other non-volatile memory devices, often supporting a mix of different storage devices. The NAS might support logical storage organization for redundancy and performance, such as mirroring and other RAID implementations -- but it's the CPU, not the disks, that handle such logical organization.
- OS. Just as with a conventional computer, the OS organizes and manages the NAS hardware and makes storage available to clients, including users and other applications. Simple NAS devices might not highlight a specific OS, but more sophisticated NAS systems might employ a discrete OS such as Netgear ReadyNAS, QNAP QTS, Zyxel FW or TrueNAS Core, among others.